

phynotype: complete workflow

phynotype provides one workflow for clustering, validation, exploration, prediction, and interpretation. This vignette is the canonical package walkthrough and replaces the older split articles.

Algorithms

K-means

Minimize

$$\min_C \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2.$$

```
fit_kmeans <- cluster(iris[, 1:4], method = "kmeans", k = 3, seed = 1)
fit_kmeans
#> <cluster_fit>
#> Method: kmeans
#> Observations: 150
#> Clusters: 3
```

PAM

Minimize the dissimilarity to medoids:

$$\min_{m_1, \dots, m_k} \sum_{i=1}^n \min_j d(x_i, m_j).$$

```
fit_pam <- cluster(iris[, 1:4], method = "pam", k = 3, seed = 1)
fit_pam
#> <cluster_fit>
#> Method: pam
#> Observations: 150
#> Clusters: 3
```

Hierarchical clustering

hclust and agnes work on numeric matrices or on dist objects created by mixed_distance().

```
d <- mixed_distance(iris[, 1:4])
fit_hclust <- cluster(d, method = "hclust", k = 3)
fit_hclust
#> <cluster_fit>
#> Method: hclust
#> Observations: 150
#> Clusters: 3
```

DBSCAN

```
fit_dbscan <- cluster(iris[, 1:4], method = "dbscan", eps = 0.5, minPts = 5)
fit_dbscan
#> <cluster_fit>
#> Method: dbscan
#> Observations: 150
#> Clusters: 2
```

Gaussian mixture models

```
fit_gmm <- cluster(iris[, 1:4], method = "gmm", k = 3, seed = 1)
fit_gmm
#> <cluster_fit>
#> Method: gmm
#> Observations: 150
#> Clusters: 3
```

K-prototypes

For mixed data, kproto combines numeric centers and categorical modes.

```
fit_kproto <- cluster(iris, method = "kproto", k = 3, seed = 1)
fit_kproto
#> <cluster_fit>
#> Method: kproto
#> Observations: 150
#> Clusters: 3
```

KMM

KMM minimizes a weighted mixed-distance criterion:

$$D(x_i, p_g) = \sum_{j \in \mathcal{N}} (x_{ij} - \mu_{gj})^2 + \lambda \sum_{j \in \mathcal{C}} I(x_{ij} \neq \nu_{gj}),$$

with $\lambda \geq 0$ controlling the categorical contribution.

```
iris_mixed <- iris
iris_mixed$Species <- as.character(iris_mixed$Species)
fit_kmm <- cluster(iris_mixed, method = "kmm", k = 3, seed = 1)
fit_kmm
#> <cluster_fit>
#> Method: kmm
#> Observations: 150
#> Clusters: 3
summary(fit_kmm)
#> Cluster fit summary
#> Method: kmm
#> Observations: 150
#> Clusters: 3
#> Sizes: 1=50, 2=50, 3=50
```

Validation metrics

The package reports silhouette width, Calinski-Harabasz, Davies-Bouldin, total within-cluster sum of squares, bootstrap ARI, ARI, and NMI.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad \text{CH} = \frac{\text{BSS}/(k-1)}{\text{WSS}/(n-k)},$$

$$\text{DB} = \frac{1}{k} \sum_{j=1}^k \max_{l \neq j} \frac{s_j + s_l}{d(\mu_j, \mu_l)}, \quad \text{NMI}(x, y) = \frac{I(x; y)}{\sqrt{H(x)H(y)}}.$$

```
validate(fit_kmeans)$metrics_table
#>      metric      value      scale      direction
#> 1 silhouette 0.5528190 -1 to 1 higher is better
#> 2 calinski_harabasz 561.6277566 positive, unbounded higher is better
#> 3 davies_bouldin 0.6619715 positive, unbounded lower is better
#> 4 total_within 78.8514414 positive, unbounded lower is better
#> 5 bootstrap_ari 0.8748382 <NA> <NA>
```

Exploration and prediction

```
exp <- explore(fit_kmeans)
head(exp$feature_summary)
#>   cluster   feature      mean      sd median min max
#> 1      1 Sepal.Length 5.901613 0.4664101    5.9 4.9 7.0
#> 2      1  Sepal.Width 2.748387 0.2962841    2.8 2.0 3.4
#> 3      1 Petal.Length 4.393548 0.5088950    4.5 3.0 5.1
#> 4      1  Petal.Width 1.433871 0.2974997    1.4 1.0 2.4
#> 5      2 Sepal.Length 6.850000 0.4941550    6.7 6.1 7.9
#> 6      2  Sepal.Width 3.073684 0.2900924    3.0 2.5 3.8
pred <- predict(fit_kmeans, iris[1:5, 1:4])
pred
#> <cluster_prediction>
#> Method: kmeans
#> Predictions: 5
```

Interpretability

`feature_importance()`, `ceteris_paribus()`, and `lime_explain()` all operate on the fitted clustering rule through `predict()`.

```
feature_importance(fit_kmeans, n_repeats = 3, seed = 1)$summary
#>      feature importance std_error n_repeats
#> 1 Petal.Length 0.53333333 0.007698004      3
#> 2 Sepal.Length 0.08222222 0.005879447      3
#> 3 Petal.Width 0.04666667 0.013877773      3
#> 4 Sepal.Width 0.01111111 0.005879447      3
```